

Bionic hand that responds automatically

A new generation of prosthetic limbs that allow the wearer to reach for objects automatically, without thinking—just like a real hand—are to be trialled for the first time. Led by biomedical engineers at Newcastle University and funded by the Engineering and Physical Sciences Research Council (EPSRC), the bionic hand is fitted with a camera which instantaneously takes a picture of the object in front of it,

assesses its shape and size, and triggers a series of movements in the hand.

Bypassing the usual processes which require the user to see the object, physically stimulate the muscles in the arm and trigger a movement in the prosthetic limb, the hand 'sees' and reacts in one fluid movement.

A small number of amputees have already trialled the new technology and now the Newcastle University team is working with experts to offer the 'hands with eyes' to patients at Newcastle's Freeman Hospital.

The findings of the study have been published in the *Journal of Neural Engineering*.

"Current prosthetic hands are controlled via myoelectric signals—that is electrical activity of the muscles recorded from the skin surface of the stump. Controlling them takes practice, concentration and, crucially, time"—says Dr Kianoush Nazarpour, co-author on the study and a senior lecturer in biomedical engineering at Newcastle University.

Using neural networks—the basis for artificial intelligence—lead author on the study Ghazal Ghazaei showed the computer numerous object images and taught it to recognise the 'grip' needed for different objects. This enables it to accurately assess and pick up an object which it has never seen before—a huge step forward in the development of bionic limbs.



Dr Kianoush Nazarpour demonstrates the bionic hand (Image courtesy: Mike Urwin/Newcastle University)

Teaching robots to teach other robots

Robots can now be made to do new types of tasks that they haven't been able to learn before, like multistep assembly using both of their arms—thanks to C-LEARN approach developed by MIT researchers.

Conventionally, most robots are programmed using one of two methods: learning from demonstration, in which they watch a task being done and then replicate it, or via motion-planning techniques such as optimisation or sampling, which require a programmer to explicitly specify a task's goals and constraints.

Both methods have drawbacks. Robots that learn from demonstration can't easily transfer one skill they've learned to another situation and remain accurate. On the other hand, motion planning systems that use sampling or optimisation can adapt to these changes but are time-consuming, since they usually have to be hand-coded by expert programmers.



MIT doctoral candidate Claudia Pérez-D'Arpino discusses her work teaching the Optimus robot to perform various tasks, including picking up a bottle (Image courtesy: MIT CSAIL)

The new approach combines the intuitiveness of learning from demonstration with the precision of motion-planning algorithms: It allows non-coders to teach robots a range